VoiceTest: Multilingual AI Collections Assistant

External Product Brief (Shareable)

1) What This Solution Does

VoiceTest automates first-line overdue EMI collection conversations through an AI voice assistant that can listen, reason, respond, and trigger downstream actions. It supports multilingual borrower interaction, captures outcomes like promise-to-pay commitments, and updates CRM/workflows in near real-time.

2) Problems It Solves

- Manual collections teams cannot sustainably cover the full overdue borrower base every day.
- Language mismatch and code-mixed conversations reduce call effectiveness and customer trust.
- Post-call operations are often delayed because CRM logging and follow-up planning are manual.
- Risk calls (disputes, distress, abuse) need immediate escalation, not batch triage.
- Leadership needs structured call intelligence (disposition + sentiment), not unstructured notes.

3) Why It Is Better (Business Value)

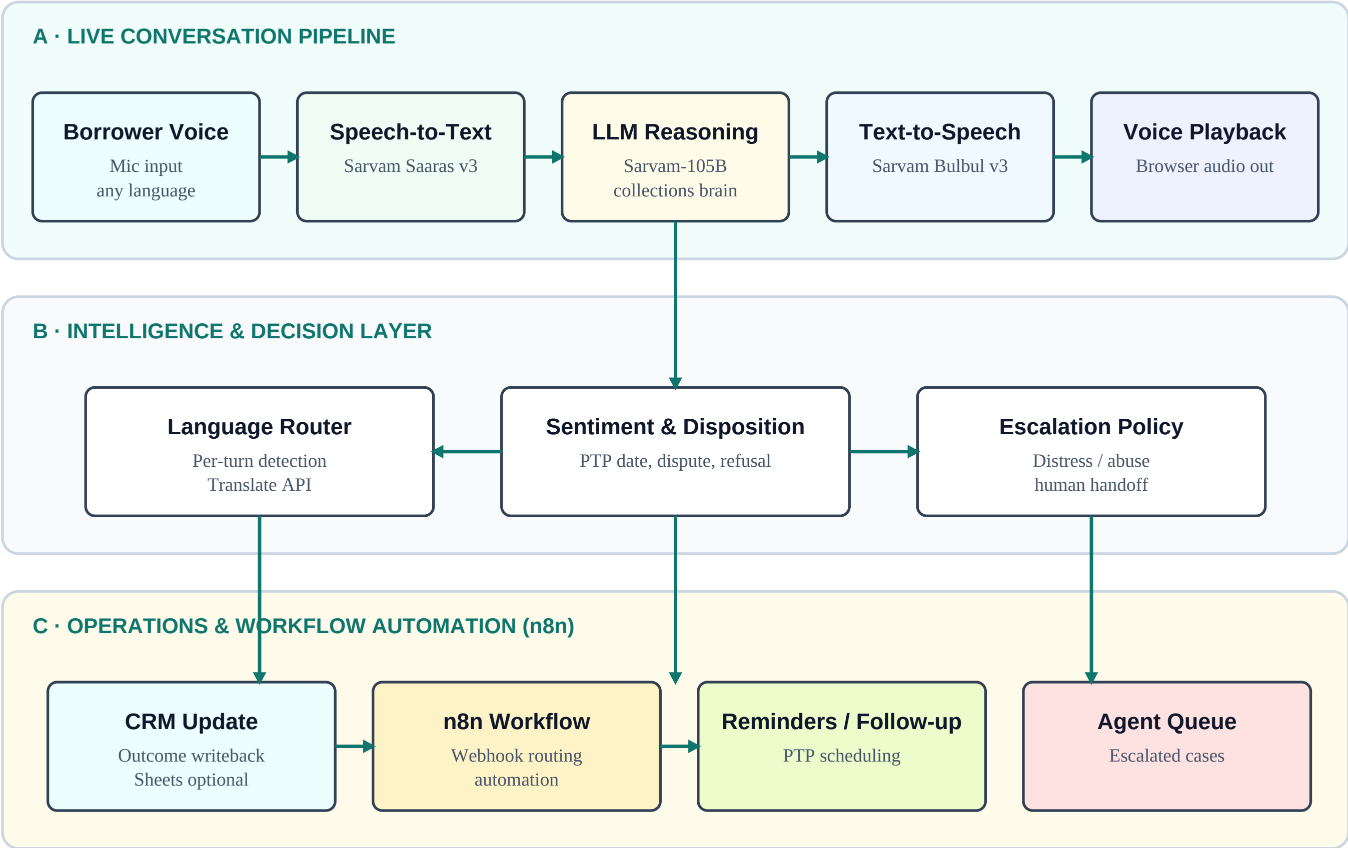
- **Cost efficiency:** AI-led first-touch interactions reduce cost per handled conversation vs. full human handling.
- **Scalable outreach:** More borrowers contacted consistently with standardized, compliance-safe messaging.
- **Operational speed:** Outcomes captured instantly; follow-ups and reminders auto-triggered via n8n.
- **Decision intelligence:** Structured sentiment/disposition signals support recovery strategy tuning.
- **Human focus:** High-risk or disputed cases escalate quickly so agents focus where expertise is needed.

4) Core Capabilities

Capability	How it works	Outcome
Multilingual interaction	Understands Hindi, Tamil, English and code-mix; adapts response language per turn.	Higher borrower engagement, lower friction.
CRM connectivity	Disposition, sentiment and payment commitment written to CRM (in-memory + optional Sheets).	Near real-time operations visibility.
Escalation mode	Dispute/distress/abuse patterns trigger escalation flags and workflow routing.	Risk calls moved to human queue quickly.
Sentiment analysis	Each turn tagged positive/neutral/negative plus call disposition.	Quantifiable call quality and strategy insight.

5) Architecture and Orchestration Flow

Real-time conversation runs left to right (top lane). Each completed turn produces structured intelligence (middle lane) that drives CRM updates, n8n automation and escalation (bottom lane).



6) Sarvam AI Components Used

Sarvam component	Role	Purpose in this solution
Saaras (saaras:v3)	Speech-to-Text	Converts borrower speech to text for turn understanding and intent capture.
Sarvam Chat (sarvam-105b)	Reasoning + response	Generates contextual collections replies and structured outcome JSON.
Bulbul (bulbul:v3)	Text-to-Speech	Produces the natural spoken reply to the borrower.
Translate API	Language control	Forces output language to match per-turn user language switching.

7) Tech Stack

- Frontend: HTML, CSS, JavaScript voice client with live conversational UX.
- Backend: Python FastAPI service for conversation handling and business logic.
- AI Services: Sarvam STT + LLM + TTS + Translate APIs.
- Workflow: n8n for post-call routing, reminders, and escalation pipelines.
- CRM/Data: In-memory CRM with optional Google Sheets integration.

GitHub Source

https://github.com/sumitJha-Realm/VoiceTest/tree/main/src

Executive Summary

VoiceTest delivers a practical, multilingual, AI-first collections workflow that combines natural borrower interaction with immediate CRM writeback and n8n orchestration. It reduces operational cost, increases outreach throughput, and preserves human intervention for high-risk conversations through escalation and sentiment-aware routing.